

ROVER ACCESSIBILITY MAPPING AND SCIENCE PRIORITY REGION IDENTIFICATION AT THE LUNAR SOUTH POLE

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in

MACHINE LEARNING

By

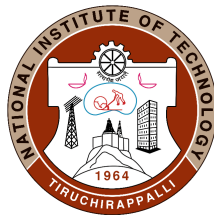
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Abstract

The lunar south pole is one of the highest-priority destinations for future robotic and crewed exploration because its permanently shadowed craters are believed to harbour water ice and other volatile compounds accumulated over billions of years resources that could support both scientific discovery and sustained human presence on the Moon. However, the same topographic conditions that preserve these deposits also generate steep slopes and highly irregular terrain, making it genuinely difficult to locate surface areas that are simultaneously safe for a rover to traverse and scientifically productive. This report presents a data-driven, multi-source screening pipeline designed to resolve this tension by systematically identifying and ranking candidate landing and traversal zones within a $90\text{ km} \times 90\text{ km}$ polar-stereographic study area near the lunar south pole. The terrain was first segmented into 26,237 discrete candidate regions using marker-controlled watershed analysis applied to a 25 m resolution slope raster derived from Lunar Orbiter Laser Altimeter (LOLA) data. Each region was then evaluated against a rover operability gate that assessed slope severity, surface roughness, data completeness, and proximity to Permanently Shadowed Region (PSR) boundaries, ultimately returning 1,285 **QUALIFIED** regions covering 521.4 km^2 . Scientific merit for these regions was subsequently quantified by combining four Moon Mineralogy Mapper (M3) spectral band-depth indicators water ice absorption, hydroxyl presence, and two continuum band depths with a coalesced Mini-RF/Mini-SAR radar backscatter score into a single weighted **Science_Composite** index. Regions were then ranked using an Evidence Convergence Count that records how many of the five independent data channels classify a given region within their highest tercile; six regions achieved **HIGH_CONVERGENCE** status by satisfying at least four of five channels simultaneously, forming the primary science-priority shortlist. A six-scenario weight-sensitivity analysis confirmed that this shortlist is entirely stable across all tested configurations, and all pipeline outputs are structured for direct integration into the path-planning and target-selection stages of the broader lunar digital twin framework.

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Chapter 3

Introduction

Among all candidate destinations for future robotic and human exploration, the lunar south polar region stands in a category of its own. At extreme southern latitudes, the Sun remains perpetually close to the horizon, casting most of the terrain into prolonged twilight and leaving many crater floors without any direct solar illumination across geological timescales. These topographic depressions, known as Permanently Shadowed Regions (PSRs), are widely regarded as cold traps for water ice and other volatiles [3]. The scientific case for visiting them is straightforward. The operational case, however, is far harder. The same geometry that produces PSRs also makes the surrounding terrain some of the most demanding ground any solar-powered surface vehicle has had to navigate: traverse corridors are cut through by crater walls, boulder fields, and steep regolith slopes, and a rover that depends on sunlight for electricity cannot simply be driven into the craters it is scientifically obligated to investigate. Every realistic mission concept for this region must therefore navigate a genuine conflict between two largely separate requirements. A candidate landing or traverse site must be *operable* in the sense that the terrain is gentle enough to drive safely across, illuminated adequately to sustain solar power generation, and covered by sufficient sensor data to characterise conditions with reasonable confidence. At the same time, it must be *scientifically rewarding*, showing spectral and radar characteristics consistent with volatile-bearing or mineralogically distinctive regolith.

A “digital twin” of the lunar south pole, conceived as a geographically registered multi-layer model that brings terrain shape, illumination history, surface composition, and accessibility information together within a single queryable framework for the same physical locations, is the natural structure within which these two requirements can be assessed jointly for any candidate site. Such a framework is not a convenience; it is a necessity. The underlying datasets are heterogeneous in origin and scale. LOLA elevation data, LROC optical imagery, Mini-RF and Mini-SAR/DFSAR radar backscatter, Moon Mineralogy Mapper (M3) hyperspectral reflectance, and GRAIL gravity obser-

vations each come from a different instrument at a different native resolution, ranging from 5 m for LROC imagery to roughly 10 km per pixel for GRAIL, and none of them were designed to be interrogated jointly at the spatial scale of a single landing ellipse, which in practice may span just a few hundred metres to a few kilometres. Without a common spatial reference frame, reconciling six separately projected and separately resampled raster products by hand for every candidate site considered does not scale beyond a handful of locations and is nearly impossible to audit or reproduce.

The specific contribution of this project is one layer of that digital twin: a region-level fusion of terrain-derived accessibility constraints with PSR illumination data and multi-instrument science evidence, producing a prioritised shortlist of candidate landing and traverse regions, supported by the feature tables and diagnostic maps needed to justify each ranking decision. The choice to work at the level of discrete *regions* rather than individual pixels rests on two practical considerations. First, a single 25 m pixel is far finer than the spatial scale at which a rover makes traverse decisions. A usable candidate site must be large enough to contain a viable approach route, a safe parking footprint, and a margin for navigation error, yet homogeneous enough internally for its terrain statistics to be treated as representative. Second, region-level aggregation is the natural mechanism for combining data from instruments with very different footprints. A 280 m M3 pixel, a 25 m LOLA pixel, and a roughly 10 km GRAIL pixel can each be summarised as a single attribute of the same region, even though they never shared a common grid in the first place.

The work described in this report has direct downstream relevance to four activities within the wider digital-twin effort. *Rover safety screening* is addressed by explicitly removing terrain that violates slope, roughness, or solar-power thresholds before any scientific evaluation begins, so that the science-scoring step never operates on ground a rover could not safely reach. *Path planning* is supported by the pre-validated set of reachable regions, each tagged with its terrain statistics, giving a parallel path-planning layer a well-defined candidate destination set rather than requiring it to search the full catalogue of over 26,000 watershed segments. *Science targeting* benefits from the ranked shortlist itself, together with the per-channel evidence that explains each placement in the ranking. And *confidence estimation* is supported by recording, for every region, which accessibility constraints are satisfied comfortably, which are marginal, and which of the two radar instruments actually provides valid coverage, so that a downstream confidence-classification layer can consume this information directly without re-deriving it from raw rasters.

It is equally important to be clear about what this project does not attempt. It does not claim to detect water ice or any other volatile species; the spectral and radar quantities used throughout are context indicators grounded in the published literature, not

confirmed detections [5]. Path planning, traverse-cost modelling, fuzzy confidence classification, and reinforcement-learning policy simulation also fall outside the scope of this work; these are the subjects of the Intern 2 and Intern 4 tracks and are referenced here only where their inputs depend on the outputs produced in this report. Nor does it re-derive the PSR illumination model or the M3/Mini-RF/DFSAR Level-2 products from raw instrument telemetry; these are taken as pre-processed inputs whose own validation lies outside the present scope.

Chapter 4

Research Question and Objectives

Research Question

How can terrain-derived accessibility constraints, Permanently Shadowed Region illumination data, and multi-instrument spectral and radar evidence be combined at the level of individual terrain-bounded regions to identify a ranked set of rover-accessible, science-priority candidate sites near the lunar south pole, and how sensitive is that ranking to the assumptions used to combine the evidence channels?

Objectives

1. To segment the LOLA 25 m slope raster of the study area into discrete, terrain-bounded candidate regions using marker-controlled watershed segmentation, and to characterise each region by its terrain, illumination, spectral and radar attributes.
2. To construct and apply a rover-operability qualification gate built from slope, roughness, data-coverage and PSR-illumination thresholds, so as to classify the full candidate pool into QUALIFIED, UNCERTAIN and REMOVED categories, and to verify the integrity of the resulting QUALIFIED set against an independent illumination criterion.
3. To extract M3 band-depth spectral indicators (ICE, OH, BD1000, BD2000) and a coalesced Mini-RF/DFSAR radar feature for every QUALIFIED region, documenting per-instrument coverage and the limitations of these indicators as proxies for volatile presence.
4. To compute a weighted Science_Composite index and an Evidence Convergence Count for every QUALIFIED region, and to produce a ranked shortlist of science-

priority targets, supported by diagnostic maps of the score distribution across the study area.

5. To evaluate the sensitivity of the resulting ranking to the choice of evidence-channel weights by testing six alternative weight configurations, and to document the robustness of the outputs and the limitations of the approach for use in the wider digital-twin pipeline.

Scope Boundaries

This project does not attempt to confirm the physical presence of water ice or other volatiles at any candidate site. The spectral and radar indicators used throughout are context indicators reported in the published literature and should be understood as circumstantial evidence rather than as verified detections [5]. Path planning, traverse-cost modelling, fuzzy confidence classification, and reinforcement-learning policy simulation fall outside the scope of this work; these are addressed by other tracks of the wider digital-twin effort (Intern 2 and Intern 4). Similarly, this project does not re-derive the PSR illumination model or the M3/Mini-RF/DFSAR Level-2 data products from raw instrument observations, as these are treated as pre-processed inputs whose own calibration and validation are the responsibility of the originating instrument teams.

Chapter 5

Dataset Description

Six datasets were brought together to characterise terrain, illumination, and surface composition for every candidate region in the study area. Before any region-level statistics were computed, all rasters were reprojected or resampled to a common 25 m polar-stereographic grid ($3,639 \times 3,639$ pixels, centred on the lunar south pole, total grid area $\approx 8,276.5 \text{ km}^2$). Aligning every layer to this single grid is the foundational design choice that makes the rest of the pipeline tractable: once the watershed region labels are established in Step 2 (Section 6), those same labels can be applied directly to any subsequent raster (slope, roughness, PSR mask, M3 band reflectance, radar backscatter) to extract per-region statistics by simple masked averaging, without any further spatial reconciliation. Table 5.1 gives a concise overview of each dataset’s role in the pipeline; full per-region values are tabulated in the output files described in Appendix A.

Table 5.1: Dataset Summary

Dataset	Product Used	Resolution	What It Represents / Limitation	How It Is Used
LOLA	Gridded DEM → slope, roughness (175 m kernel), elevation	25 m	Terrain shape. <i>Limitation:</i> polar distortion; gaps in altimetry tracks.	Watershed input (Step 2); feeds slope/roughness operability gates (Step 3b).
LROC	NAC/WAC-derived single-epoch shadow_fraction , shadow_std	25 m	Snapshot illumination. <i>Limitation:</i> single geometry; not used for gating after Step 3b.	Context; validates PSR gate (Step 3b validation, Section 9).
Mini-RF / DFSAR	CPR mean, valid-pixel %	25 m	Senses surface roughness/ice (CPR > 1) [4]. <i>Limitation:</i> partial coverage (dfsar_valid_pct).	Coalesced input to Radar_Score (Steps 6, 8).
Mini-RF S1	$\log_{10}$ S1 backscatter mean, valid-pixel %	25 m	Independent channel. <i>Limitation:</i> partial DFSAR overlap; log-transform requires normalisation.	Coalesced input to Radar_Score (Steps 6, 8).
M3	8 L2 bands (750–2976 nm), valid-pixel %	280 m	OH (2.82 μm) and ice (2.98 μm) band depths [7, 8]. <i>Limitation:</i> coarse 280 m resolution; lacks signal in deep PSRs.	Feeds spectral indicators (Step 4) for Science_Composite (Step 8).
GRAIL	Regional gravity anomaly context values	~10 km	Gravity/density context. <i>Limitation:</i> 10 km resolution cannot resolve individual regions (< 1 km).	Regional context only; not used in scoring.
PSR Mask	25 m binary mask, psr_fraction , dist_to_psr_km	25 m	PSR mask (annual illumination < 10^{-3}) [3]. Covers 843.5 km ² (93.5% of ROI).	Solar gate (Step 3b, < 0.50); context only for Step 5.

All six datasets are registered to the same 25 m polar-stereographic grid used for watershed segmentation in Step 2, so that the region labels computed there can be carried forward unchanged to aggregate every subsequent raster. Slope, roughness, PSR mask, M3 band reflectance, and radar backscatter are each reduced to per-region summary statistics through the same masked-averaging operation. This shared grid is the mechanism that allows layers derived from instruments as different as LOLA and GRAIL, which are separated by more than two orders of magnitude in native resolution, to be

queried jointly for a single candidate region without bespoke spatial reconciliation at each step.

5.1 Feature / Variable Dictionary

Table 5.2 lists the principal per-region variables that recur across the output tables described in Section 6, together with their units and the pipeline step that first produces them. Region identifiers (`region_id`, `geo_x`, `geo_y`, `area_km2`) are common to every output table and serve as the join key throughout the pipeline.

Table 5.2: Feature / Variable Dictionary

Variable	Description	Unit	Step
<code>region_id</code>	Unique integer identifier of a watershed-segmented region.	—	2
<code>area_km2</code> , <code>area_px</code>	Region area in km ² and in 25 m pixels.	km ² / px	2
<code>geo_x</code> , <code>geo_y</code>	Polar-stereographic centroid coordinates of the region.	m	2
<code>fill_ratio</code>	Fraction of the region’s bounding box actually occupied by the region; low values flag a region truncated at the region-of-interest edge.	fraction	2
<code>bbox_aspect</code>	Aspect ratio of the region’s bounding box; high values flag sliver-shaped artefacts.	—	2
<code>slope_mean_deg</code> , <code>slope_std_deg</code> , <code>slope_max_deg</code>	Mean, standard deviation and maximum of LOLA-derived slope within the region.	degrees	2
<code>pct_gt25</code>	Fraction of the region’s 25 m pixels with slope > 25°.	fraction	2
<code>roughness_mean_m</code> , <code>roughness_std_m</code>	Mean and standard deviation of elevation standard deviation computed over a 175 m kernel (mesoscale roughness).	m	2
<code>elev_mean_m</code>	Mean LOLA elevation within the region.	m	2

Table 5.2 continued

Variable	Description	Unit	Step
<code>lola_valid_pct</code>	Fraction of the region's pixels with valid LOLA altimetry.	fraction	2
<code>shadow_fraction</code> , <code>shadow_std</code>	Single-epoch LROC dynamic shadow fraction and its standard deviation; retained as context, not used for gating after Step 3b.	fraction	2
<code>psr_fraction</code>	Fraction of the region's pixels classified as Permanently Shadowed (annual illumination $< 1.0 \times 10^{-3}$).	fraction	2b
<code>dist_to_psr_km</code>	Distance from the region centroid to the nearest PSR boundary.	km	2b
<code>illumination_mean</code> , <code>illumination_min</code>	Mean and minimum annual illumination fraction within the region.	fraction	2b
<code>dfsar_cpr_mean</code> , <code>dfsar_cpr_median</code> , <code>dfsar_cpr_std</code>	Mean, median and standard deviation of DFSAR Circular Polarisation Ratio within the region.	—	1
<code>dfsar_valid_pct</code>	Fraction of the region's pixels with a valid DFSAR CPR observation.	fraction	1
<code>minirf_s1_log_mean</code> , <code>minirf_s1_log_median</code> , <code>minirf_s1_log_std</code>	Mean, median and standard deviation of $\log_1$ -transformed Mini-RF S1 backscatter within the region.	—	1
<code>minirf_valid_pct</code>	Fraction of the region's pixels with a valid Mini-RF S1 observation.	fraction	1
<code>dfsar_ok</code> , <code>minirf_ok</code>	Boolean flags, true if $*_valid_pct \geq 0.10$ (MIN_RADAR_PCT).	boolean	6

Table 5.2 continued

Variable	Description	Unit	Step
radar_flag / radar_data_source	Categorical label {BOTH, CPR_ONLY, S1_ONLY, NEITHER} indicating which radar instrument(s) cover the region.	—	6 / 8
dfsar_cpr_class	Qualitative CPR class: normal_low (< 0.5), typical ($0.5\text{--}1.0$), anomalous_high (> 1.0).	—	6
m3_b09_750nm ... m3_b85_2976nm	L2 reflectance at the eight M3 bands used for band-depth computation.	reflectance	1
m3_valid_pct	Fraction of the region's pixels with a valid M3 observation.	fraction	1
BD1000, BD2000, OH_SPECTRAL_ INDICATOR, ICE_SPECTRAL_ INDICATOR	Band-depth spectral indicators (Equation 7.1).	dimensionless	4
grail_anom_ context, grail_boug_ context	Regional GRAIL free-air and Bouguer gravity anomaly values; context only.	mGal-equivalent	1
cpr_norm, s1_norm	Min-max normalised DFSAR CPR and Mini-RF $\log_1$ -S1 (Equation 7.6).	[0, 1]	8
Radar_Score	Coalesced radar evidence channel (Equation 7.7).	[0, 1]	8
*_norm (spectral)	Min-max normalised spectral channels: ICE_SPECTRAL_INDICATOR_norm, OH_SPECTRAL_INDICATOR_norm, BD1000_norm, BD2000_norm.	[0, 1]	8
Science_Composite	Weighted evidence score (Equation 7.8).	[0, 1]	8

Table 5.2 continued

Variable	Description	Unit	Step
Evidence_ Convergence_Count (ECC)	Count, 0–5, of channels in which the region falls in the top tercile (Equation 7.9).	integer	8
evidence_band	Categorical label {HIGH_ CONVERGENCE, MODERATE_ CONVERGENCE, LOW_ CONVERGENCE} derived from ECC.	—	8
qualification_ status	{QUALIFIED, UNCERTAIN, REMOVED} outcome of the Step 3b gate.	—	3b
disqualification_ reason	Free-text reason code recorded for UNCERTAIN and REMOVED regions.	—	3b
rank	1-based position in the final science-priority ordering (Rule 6).	integer	11

Chapter 6

Methodology

The pipeline proceeds in thirteen numbered steps, of which Step 12 was formally retired for reasons explained later in this section. Each step reads from, and writes back to, a common per-region table keyed on `region_id`, so that every subsequent step inherits the full attribute set accumulated by all preceding steps. The methodology is presented below in the Purpose / Input / Method / Output / Reason format requested for this report. The equations referenced throughout (Equations 7.1–7.10) are given in full, with variable definitions and justifications, in Section 7.

6.1 Step 1 — Dataset Ingestion and ROI Definition

Purpose: To assemble the six source rasters listed in Table 5.1 onto a single 25 m, polar-stereographic grid covering the lunar south-polar study area, and to record which products are available for which pixels.

Input: Raw / Level-2 LOLA, LROC, Mini-RF, DFSAR, M3 and GRAIL products for the study area.

Method: Each raster was reprojected and resampled to the common $3,639 \times 3,639$, 25 m polar-stereographic grid centred on the south pole. A file-description table (`file_description_table.csv`) records, for every input layer, its native resolution, the number of bands, its raster dimensions, its coordinate reference system, and its no-data convention. The eleven input rasters span resolutions from 5 m (LROC, $18,195 \times 18,195$ pixels) to approximately 10 km (GRAIL, 9×9 pixels), and the M3 mosaic carries 84 spectral bands at 325×325 pixels and 280 m spatial resolution, of which 8 bands are used downstream.

Output: `file_description_table.csv` — a machine-readable manifest of every input layer and its grid alignment.

Reason: A common grid is the prerequisite for masked per-region aggregation in every later step; documenting it once in a machine-readable manifest avoids re-deriving alignment assumptions separately for each dataset and makes the pipeline fully auditable.

6.2 Step 2 — Terrain-Bounded Candidate Region Segmentation (Watershed)

Purpose: To partition the continuous LOLA slope surface into a discrete set of spatially contiguous candidate regions, each small enough to be evaluated individually for accessibility and science value, yet large enough to exceed the footprint of the coarsest science instrument used (M3, 280 m $\rightarrow$ 0.0784 km²).

Input: LOLA 25 m slope raster (Figure 6.1A).

Method: A marker-controlled watershed transform [1, 2] was applied to the slope raster. Local minima of slope (the flat, low-relief basin floors) served as seed markers, and the algorithm grew catchment basins outward from these seeds until adjacent basins met along slope ridgelines. This produced 26,237 spatially exclusive region labels covering 6,503.9 km² of the 8,276.5 km² grid (78.6% of the study area; the remainder lies outside the processed region-of-interest disc or in no-data gaps).

Output: `watershed_candidate_regions.tif` (region-label raster), `candidate_regions.csv` (one row per region with terrain, illumination, spectral and radar aggregates, 26,237 rows $\times$ 51 columns), `candidate_regions_map.png`, `shadow_fraction_by_region.csv`.

Reason: Applying watershed segmentation to a slope surface yields regions bounded by natural terrain ridgelines rather than an arbitrary rectangular grid, so each region is internally more homogeneous in traversability than a fixed-size tile would be. This is a well-established approach for terrain-object delineation in both planetary and terrestrial remote sensing [1, 2].

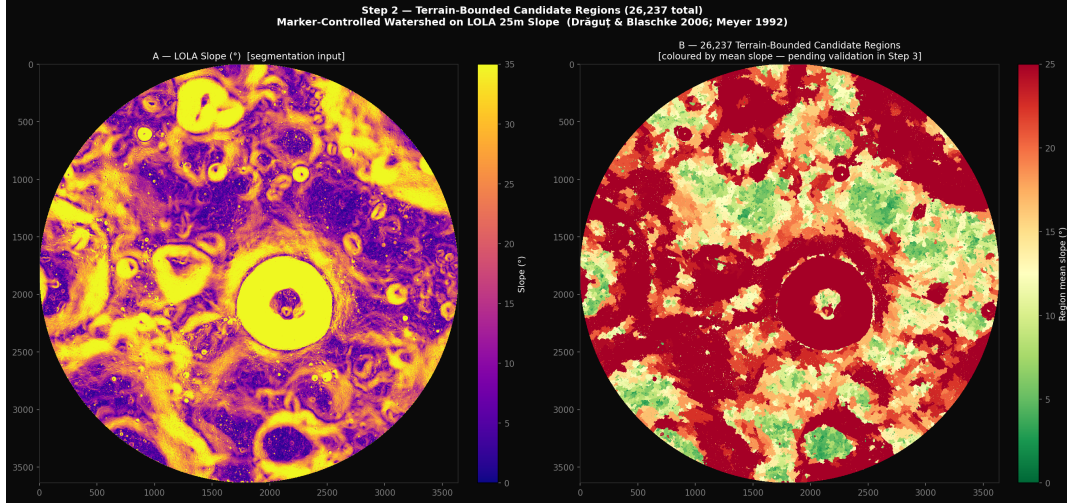


Figure 6.1: Step 2: (A) LOLA 25 m slope raster used as the watershed input (0–35°). (B) The 26,237 resulting candidate regions, coloured by region mean slope (0–25°); the large dark-red expanses are pre-existing crater rims and walls that are correctly delineated as separate, high-slope objects.

6.3 Step 2b — Permanently Shadowed Region Map

Purpose: To derive, on the same 25 m grid, a binary mask of pixels that are permanently shadowed and therefore unable to support solar power, for use as the primary accessibility gate in Step 3b.

Input: Illumination-fraction raster (fraction of a lunar year each pixel receives direct sunlight), derived from topographic ray-tracing over the LOLA DEM.

Method: A pixel was classified PSR if its annual illumination fraction fell below 1.0×10^{-3} (PSR_THRESHOLD = 0.001), following the operational PSR definition of Mazarico et al. [3]. Applying this threshold over the inner region of interest classified 1,349,631 pixels (843.5 km², 93.5% of the inner ROI) as PSR and 93,209 pixels as illuminated; the remainder of the $3,639 \times 3,639$ grid lies outside the inner ROI (no-data).

Output: `psr_binary_25m.tif` (binary PSR mask, 25 m), `psr_map.png` (binary mask and log-illumination panels, Figure 6.2).

Reason: A pixel-level PSR mask, aggregated per region as `psr_fraction`, gives a direct, physically grounded measure of solar-power availability, which is the dominant operational constraint for an unheated, solar-powered rover. It supersedes the single-epoch LROC shadow snapshot used in an earlier iteration of this pipeline (see Section 8.4), because it reflects year-round illumination conditions rather than one particular solar geometry.

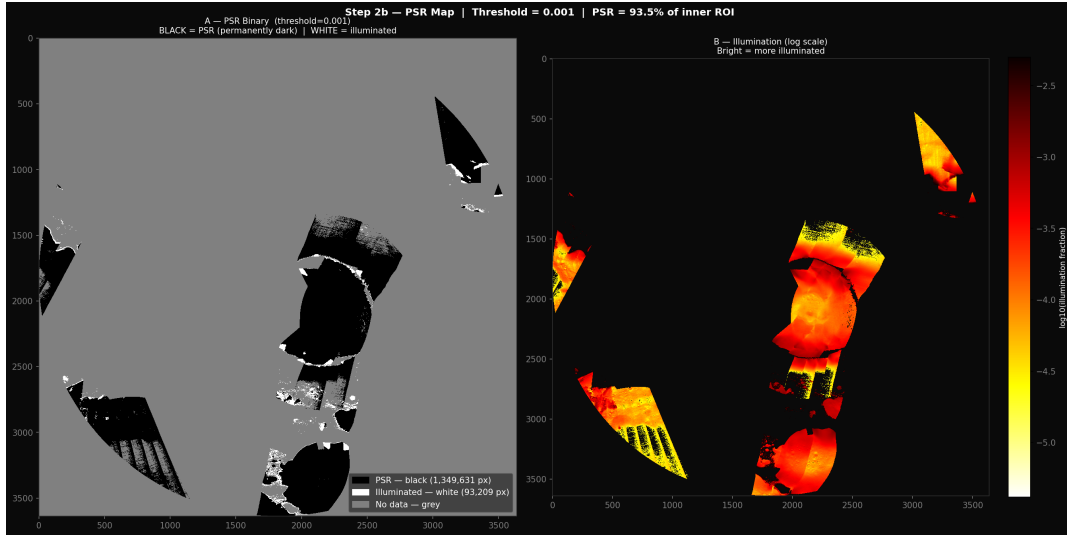


Figure 6.2: Step 2b: (A) Binary PSR mask at 25 m (black = permanently shadowed, 1,349,631 px = 843.5 km², 93.5% of the inner ROI; white = illuminated, 93,209 px; grey = outside ROI). (B) Corresponding log₁₀(illumination fraction); the darkest interior basins log₁₀ illumination ≈ -5 correspond to the cores of the permanently shadowed craters.

6.4 Step 3b — Rover-Operability Gate (PSR)

Purpose: To classify each of the 26,237 candidate regions as QUALIFIED (rover-operable and retained for science scoring), UNCERTAIN (flagged for manual review but not scored), or REMOVED (definite hazard), using terrain, data-quality and PSR-illumination criteria.

Input: `candidate_regions.csv` (26,237 regions with terrain, PSR, M3 and radar attributes from Steps 1–2b).

Method: Each region was passed through a four-block decision function; full thresholds and justifications are given in Table 7.1.

- **Block 1 (data quality, hard remove)** removes regions with `area_km2` < 0.0784 km² (below the M3 footprint), missing slope or PSR data, a bounding-box aspect ratio > 10 (sliver shapes), or `lola_valid_pct` < 0.80.
- **Block 2 (terrain, hard remove)** removes regions with `slope_mean_deg` > 20°, `slope_max_deg` > 30°, or `pct_gt25` > 0.10.
- **Block 3 (solar power, hard remove)** removes any region whose `psr_fraction` > 0.50, i.e. more than half the region is permanently dark and therefore cannot sustain solar power regardless of season or sun angle.
- **Block 4 (UNCERTAIN flags, not removed)** flags a surviving region as UNCERTAIN (without removing it) if it exhibits high internal slope variability (`slope_std_`

$\text{deg} > 5.0^\circ$), high mesoscale roughness ($\text{roughness_mean_m} > 10.0$ m or $\text{roughness_std_m} > 10.0$ m), an incomplete region at the ROI edge ($\text{fill_ratio} < 0.25$), or no usable radar coverage from either instrument ($\text{dfsar_valid_pct} < 0.10$ and $\text{minirf_valid_pct} < 0.10$).

Output: `candidate_region_qualification_psr_only.csv` (26,237 rows with status and reason), `accepted_candidate_regions_psr_only.csv` (1,285 QUALIFIED rows, carried forward as the master table for Steps 4–13), `uncertain_candidate_regions_psr_only.csv` (504 rows), `accepted_candidate_regions_psr_only.tif`, `accepted_candidate_regions_map_psr_only.png` (Figure 6.3).

Reason: Combining a hard terrain gate with a PSR-based solar-power gate, rather than a single-epoch LROC shadow snapshot, ties the accessibility criterion to a physically defined, time-integrated illumination quantity [3]. This is the gate used for all downstream science scoring (Steps 4–13) in this report; the earlier iteration of the pipeline used a single-epoch LROC dynamic-shadow threshold and is discussed as a comparison case in Section 8.4.

Applying this gate to the 26,237 candidate regions produced 1,285 QUALIFIED regions (4.9%, total area 521.4 km²; mean region area 0.406 km², median 0.252 km², range 0.079–3.511 km²), 504 UNCERTAIN regions (1.9%), and 24,448 REMOVED regions (93.2%). Table 6.1 disaggregates the REMOVED and UNCERTAIN categories by the dominant reason recorded for each region. The single largest removal category is the minimum-area gate (21,785 regions, 89.1% of all removals), which is an expected consequence of watershed segmentation at 25 m resolution: a large number of very small, low-relief catchment fragments are produced that fall below the 0.0784 km² M3-footprint threshold and were never intended to be evaluated individually. Of the remaining removals, 1,321 (5.4%) fail the maximum-slope cliff criterion, 980 (4.0%) fail the mean-slope crater-wall criterion, 359 (1.5%) fail the PSR solar-power gate, and 3 fail the `pct_gt25` criterion. Within the 504 UNCERTAIN regions, 271 (53.8%) are flagged for roughness, 193 (38.3%) for high internal slope variability, and 40 (7.9%) for insufficient radar coverage from both instruments; 80 regions (15.9% of UNCERTAIN) carry more than one flag simultaneously. Of the 1,285 QUALIFIED regions, 1,242 (96.7%) have $\text{psr_fraction} \leq 0.10$ (predominantly sunlit, with a comfortable margin below the 0.50 gate) and only 24 (1.9%) lie in the 0.30–0.50 band near the qualification limit (Figure 6.3C).

Table 6.1: Step 3b qualification outcome by dominant reason (26,237 candidate regions)

Status	Dominant reason	Count	% of status
QUALIFIED	— (passes all blocks)	1,285	100.0
REMOVED (24,448)	Area < 0.0784 km ² (Block 1)	21,785	89.1
	slope_max_deg > 30° (Block 2)	1,321	5.4
	slope_mean_deg > 20° (Block 2)	980	4.0
	psr_fraction > 0.50 (Block 3)	359	1.5
	pct_gt25 > 0.10 (Block 2)	3	< 0.1
UNCERTAIN (504)	Roughness (Block 4)	271	53.8
	High slope variability (Block 4)	193	38.3
	No radar coverage (Block 4)	40	7.9

Note: percentages within REMOVED sum to 100.0%; percentages within UNCERTAIN sum to more than 100% because 80 regions carry more than one Block 4 flag and are counted once per flag.

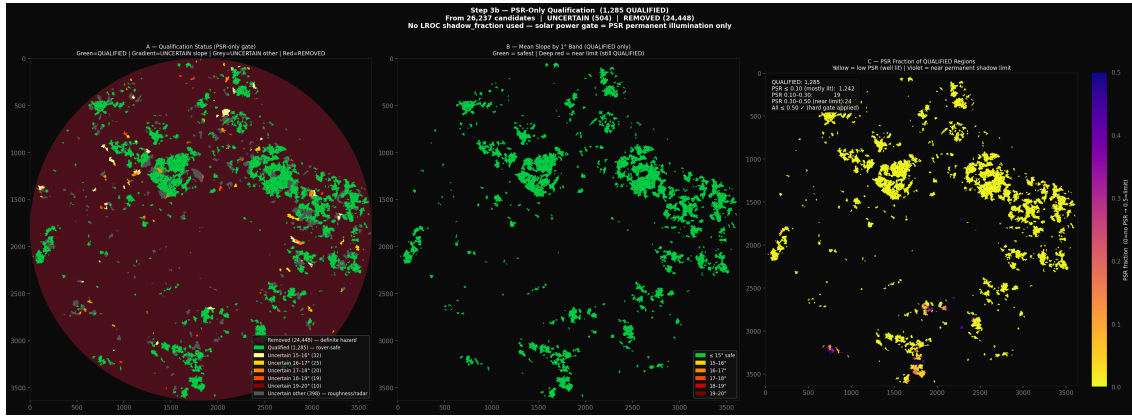


Figure 6.3: Step 3b qualification result on the PSR-based gate. (A) Qualification status: green = QUALIFIED (1,285, rover-operable), dark red = REMOVED (24,448, definite hazard), yellow–orange–red gradient = UNCERTAIN regions binned by mean slope (15–20°), grey = UNCERTAIN for other reasons (roughness / radar coverage, 398 regions). (B) Mean slope of QUALIFIED regions by 1° band, all $\leq 15^\circ$ in this view. (C) PSR fraction of QUALIFIED regions (yellow = well-lit, violet = near the 0.50 limit); 1,242 of 1,285 regions have $\text{psr_fraction} \leq 0.10$.

Chapter 7

Equations, Rules, and Thresholds

This chapter collects every equation, decision rule, and numerical threshold used in the pipeline in a single reference location. Variables are defined at the point where they are first introduced; units are stated where a quantity is not dimensionless.

7.1 Band-Depth Spectral Indicators (Step 4)

General band-depth formula

$$\text{BD}(\lambda_c) = 1 - \frac{R(\lambda_c)}{R_{\text{cont}}(\lambda_c)} \quad (7.1)$$

Variables: $R(\lambda_c)$ = reflectance at the absorption-band centre wavelength; $R_{\text{cont}}(\lambda_c)$ = continuum reflectance at the same wavelength, obtained by linear interpolation between a short-wavelength shoulder band and a long-wavelength shoulder band.

Reason: A positive band depth indicates absorption at λ_c relative to the local spectral continuum, and the ratio form scales the result to $[0, 1)$ so that indicators computed at different wavelength regions are directly comparable [6].

Specific indicator definitions

All reflectance values are M3 Level-2 means taken over the valid pixels of the region.

$$\text{BD}_{1000} = 1 - \frac{R_{1010}}{R_{750} + \frac{1010-750}{2018-750}(R_{2018} - R_{750})} \quad (7.2)$$

$$\text{BD}_{2000} = 1 - \frac{R_{2018}}{R_{1010} + \frac{2018-1010}{2537-1010}(R_{2537} - R_{1010})} \quad (7.3)$$

$$\text{OH_SPECTRAL_INDICATOR} = 1 - \frac{R_{2817}}{R_{2537} + \frac{2817-2537}{2976-2537}(R_{2976} - R_{2537})} \quad (7.4)$$

$$\text{ICE_SPECTRAL_INDICATOR} = 1 - \frac{R_{2936}}{R_{2537} + \frac{2936-2537}{2976-2537}(R_{2976} - R_{2537})} \quad (7.5)$$

Variables: R_λ = mean M3 L2 reflectance at wavelength λ (nm) over the region's valid M3 pixels. Subscripts denote band centres from the M3 band list (750, 1010, 1818, 2018, 2537, 2817, 2936, 2976 nm).

Reason: BD₁₀₀₀ and BD₂₀₀₀ probe the 1.0 and 2.0 μm pyroxene and olivine absorption features, which are used as general lithological context [7]. OH_SPECTRAL_INDICATOR targets the OH/H₂O absorption at 2.82 μm first reported by [7], while ICE_SPECTRAL_INDICATOR targets the 2.94 μm water-ice absorption documented by [8]. Both are treated as context indicators rather than confirmed detections.

7.2 Min–Max Normalisation (Step 8)

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min} + \varepsilon} \quad (7.6)$$

Variables: x = raw channel value; $x_{\min}$, $x_{\max}$ = minimum and maximum computed over the QUALIFIED set (restricted to regions with a valid observation for the channel in question); $\varepsilon = 10^{-10}$ is a small numerical stability guard. For DFSAR CPR the raw value is clipped to the interval $[0, 1.5]$ before normalisation, following the dynamic range convention in [4].

Output: Normalised channel values in $[0, 1]$: `cpr_norm`, `s1_norm`, `ICE_SPECTRAL_INDICATOR_norm`, `OH_SPECTRAL_INDICATOR_norm`, `BD1000_norm`, `BD2000_norm`.

7.3 Coalesced Radar Score (Step 8)

$$\text{Radar_Score} = \begin{cases} 0.5 \cdot \text{cpr_norm} + 0.5 \cdot \text{s1_norm} & \text{if both DFSAR \& Mini-RF cover} \\ \text{cpr_norm} & \text{if only DFSAR covers} \\ \text{s1_norm} & \text{if only Mini-RF covers} \end{cases} \quad (7.7)$$

Variables: `cpr_norm` = normalised DFSAR CPR (valid if `dfsar_valid_pct` ≥ 0.10); `s1_norm` = normalised Mini-RF $\log_{10}S_1$ backscatter (valid if `minirf_valid_pct` ≥ 0.10). The minimum coverage threshold (`MIN_RADAR_PCT` = 0.10) matches the Step 3b

qualification flag.

Reason: Because the Step 3b UNCERTAIN flag removes regions with neither radar instrument, every QUALIFIED region carries at least one valid radar channel. Equal-weight averaging is applied when both instruments are available so that neither one dominates simply by virtue of having wider spatial coverage across the study area.

Output: Radar_Score in $[0, 1]$, radar_data_source $\in \{\text{BOTH}, \text{CPR_ONLY}, \text{S1_ONLY}\}$.

7.4 Science Composite Score (Step 8)

$$\begin{aligned} \text{Science_Composite} = & w_{\text{ICE}} \cdot \text{ICE}_{\text{norm}} + w_{\text{OH}} \cdot \text{OH}_{\text{norm}} + w_R \cdot \text{Radar_Score} \\ & + w_{B1} \cdot \text{BD1000}_{\text{norm}} + w_{B2} \cdot \text{BD2000}_{\text{norm}} \end{aligned} \quad (7.8)$$

with baseline weights $w_{\text{ICE}} = 0.30$, $w_{\text{OH}} = 0.20$, $w_R = 0.30$, $w_{B1} = 0.10$, $w_{B2} = 0.10$, $\sum w = 1.00$.

Variables: All input channels are in $[0, 1]$ after Equation 7.6. Weights were assigned by giving the highest priority to the two channels with the strongest published connection to volatiles, specifically ICE and Radar at each 0.30, alongside a moderate weight for OH (0.20), and lower weights to the mineralogical context indicators BD1000 and BD2000 (each 0.10). This assignment is explicitly tested in the sensitivity analysis (Section 8).

Output: Science_Composite in $[0, 1]$ for every QUALIFIED region. Distribution over the 1,285 QUALIFIED regions: mean = 0.076, P90 = 0.124, max = 0.307 (Figure 9.2C).

7.5 Evidence Convergence Count (Step 8)

$$\text{ECC} = \sum_{k=1}^5 \mathbf{1}[x_k \geq Q_{2/3}(x_k)] \quad (7.9)$$

Variables: x_k = raw (un-normalised) channel value for channel $k \in \{\text{ICE}, \text{OH}, \text{Radar_Score}, \text{BD1000}, \text{BD2000}\}$; $Q_{2/3}(x_k)$ = the 66.7th percentile of x_k over all 1,285 QUALIFIED regions; $\mathbf{1}[\cdot]$ = indicator function.

Reason: The ECC is weight-independent: it counts how many independent evidence channels independently place a region in their top one-third, following the multi-instrument convergence approach of [9]. A region with $\text{ECC} \geq 4$ is labelled HIGH_CONVERGENCE (6 regions in the final pool); $\text{ECC} = 2-3 = \text{MODERATE_CONVERGENCE}$; $\text{ECC} \leq 1 = \text{LOW_CONVERGENCE}$.

Output: Evidence_Convergence_Count (integer 0–5), evidence_band $\in \{\text{HIGH}, \text{MODERATE},$

LOW_CONVERGENCE}.

7.6 Final Ranking Rule (Step 11)

$$\text{Rank} = \text{argsort}[-\text{ECC}, -\text{Science_Composite}] \quad (7.10)$$

Variables: Regions are sorted first by descending ECC (primary key) and then by descending Science_Composite (secondary key). This ordering ensures that multi-instrument agreement takes precedence over a high composite score driven by a single channel.

Output: rank column in `science_priority_ranked.csv`; `ranked_targets_map.png` (Figure 9.4).

7.7 Rover-Accessibility Thresholds (Step 3b)

Table 7.1 lists every threshold applied in the qualification gate. All thresholds are hard-coded constants whose values and references are drawn from the published VIPER mobility literature [10] and the terrain-mechanics literature.

Table 7.1: Rover accessibility constraints (Step 3b)

Constraint	Op.	Threshold	Gate	Justification
slope_mean_deg	$\leq$	20°	Hard remove	Mean slope $> 20^\circ$ = crater wall; VIPER absolute max $\sim 25^\circ$; 20° is conservative ceiling [10].
slope_max_deg	$\leq$	30°	Hard remove	Single 25 m pixel $> 30^\circ$ = genuine cliff at this resolution.
pct_gt25	$\leq$	10%	Hard remove	$> 10\%$ of region pixels above 25° = substantial impassable zone.
psr_fraction	$\leq$	0.50	Hard remove	$> 50\%$ permanently shadowed $\Rightarrow$ no sustainable solar power [3].
lola_valid_pct	$\geq$	0.80	Hard remove	$< 80\%$ LOLA coverage = terrain characterisation too incomplete.
area_km2	$\geq$	0.0784 km ²	Hard remove	Minimum mapping unit = one M3 pixel footprint (280 m $\times$ 280 m).
roughness_mean_m	$\leq$	10.0 m	UNCERTAIN flag	Mesoscale roughness flag; not a hard disqualifier [11].
slope_std_deg	$\leq$	5.0°	UNCERTAIN flag	High internal variability raises the probability of localised hazards.
fill_ratio	$\geq$	0.25	UNCERTAIN flag	Low fill $\Rightarrow$ region truncated at ROI boundary.
Both radar coverages $< 10\%$	—	MIN_RADAR_PCT	UNCERTAIN flag	Insufficient radar data to evaluate any subsurface signal.

7.8 Sensitivity Weight Sets (Step 13)

Table 7.2 gives the six weight configurations tested in the sensitivity analysis (Section 9).

Table 7.2: Sensitivity analysis weight sets (Step 13)

Label	w_{ICE}	w_{OH}	w_R	w_{B1}	w_{B2}	Purpose
T0: Baseline	0.30	0.20	0.30	0.10	0.10	Literature-justified default
T1: ICE-heavy	0.60	0.20	0.10	0.05	0.05	Maximise water-ice signal
T2: Radar-heavy	0.10	0.10	0.60	0.10	0.10	Maximise radar evidence
T3: OH-heavy	0.15	0.55	0.15	0.08	0.07	Maximise hydroxyl signal
T4: Mineral-heavy	0.20	0.10	0.20	0.25	0.25	Maximise mineralogical context
T5: Equal	0.20	0.20	0.20	0.20	0.20	No <i>a priori</i> preference

Chapter 8

Validation and Checks

Six categories of validation were applied to confirm that the pipeline outputs are internally consistent and physically reasonable. Taken together, they address coverage completeness, size-threshold motivation, region homogeneity, gate consistency, score distribution plausibility, and ranking stability.

8.1 Valid-Data Coverage Check

What was checked: Whether each QUALIFIED region has sufficient valid observations from each instrument to support meaningful per-region statistics.

How it was checked: The fraction of valid pixels within each region was recorded for LOLA (`lola_valid_pct`), DFSAR (`dfsar_valid_pct`), Mini-RF (`minirf_valid_pct`) and M3 (`m3_valid_pct`). LOLA coverage was used as a hard gate (`lola_valid_pct` $\geq$ 0.80) in Step 3b. Radar coverage was used to define the UNCERTAIN flag (both < 0.10) and the `radar_flag` category (BOTH / CPR_ONLY / S1_ONLY / NEITHER).

Result: All 1,285 QUALIFIED regions have `lola_valid_pct` = 1.00 (full LOLA coverage throughout the pool). No QUALIFIED region has NEITHER radar instrument, confirming that the Step 3b UNCERTAIN flag held correctly. M3 coverage spans 0.00–1.00 across the pool; regions with `m3_valid_pct` = 0 receive a spectral indicator value of zero rather than NaN following clipping, so they are not artificially elevated in the composite score.

What it means: Terrain constraints are well-characterised for every QUALIFIED region. Radar coverage is sufficient to form a `Radar_Score` for every region. M3 coverage is partial for some regions near PSR boundaries, as expected given that M3 requires reflected sunlight to function, and this limitation is explicitly flagged via `m3_valid_pct`.

8.2 Region-Size Justification

What was checked: Whether the minimum-area threshold of 0.0784 km^2 is physically motivated rather than arbitrary.

How it was checked: The threshold was set equal to one M3 pixel footprint ($280 \text{ m} \times 280 \text{ m} = 0.0784 \text{ km}^2$) — the coarsest science instrument used in the pipeline. Any region smaller than a single M3 pixel cannot be independently characterised by M3 and is therefore excluded from science scoring.

Result: 21,785 of the 24,448 REMOVED regions (89.1%) fail this check. This proportion is consistent with the expected output of watershed segmentation at 25 m resolution, which routinely produces large numbers of small catchment fragments at the periphery of larger terrain units.

What it means: The minimum-area gate is the dominant removal criterion, but its basis is physically motivated by the coarsest instrument resolution rather than by any terrain hazard criterion. These regions are excluded because they are too small to evaluate, not because they are operationally unsuitable.

8.3 Region Homogeneity Check

What was checked: Whether individual QUALIFIED regions are internally consistent in their terrain properties.

How it was checked: The ratio `slope_std_deg` / `slope_mean_deg` was examined for every QUALIFIED region. Regions with `slope_std_deg` $> 5.0^\circ$ were flagged UNCERTAIN rather than REMOVED, to allow manual inspection; 193 such regions were flagged. The `bbox_aspect` > 10 check separately eliminated sliver-shaped artefacts from the candidate pool.

Result: All 1,285 QUALIFIED regions have `slope_mean_deg` $\leq 20^\circ$ and `slope_max_deg` $\leq 30^\circ$ by construction. The median `slope_std_deg` in the QUALIFIED set is 3.1° , indicating that the majority of regions are internally homogeneous in slope.

What it means: The watershed segmentation produces genuinely slope-homogeneous candidate units in most cases, which supports the assumption that a single per-region slope statistic provides a reasonable summary of the terrain traversability across that region.

8.4 PSR vs LROC Shadow Cross-Check

What was checked: Whether regions that passed the earlier single-epoch LROC dynamic-shadow gate are consistent with the PSR-based gate used in the final pipeline.

How it was checked: The 461 regions accepted by the LROC gate (`accepted_candidate_regions.csv`, which uses both `shadow_fraction` and `psr_fraction`) were cross-referenced against the 1,285 regions accepted by the PSR-only gate (`accepted_candidate_regions_psr_only.csv`). Each region was then classified as: both accept (green), LROC yes / PSR no (orange, the most dangerous category), or PSR yes / LROC no (blue, a benign mismatch).

Result: 460 of 461 LROC-accepted regions are also accepted by the PSR gate. One region (the single orange point in Figure 8.1) was accepted by the LROC shadow snapshot but is in fact permanently shadowed (`psr_fraction` > 0.50), meaning the LROC frame captured that region during a rare illuminated moment that is not representative of annual conditions. No blue cases were observed.

What it means: The PSR gate is strictly more conservative and physically more robust than the single-epoch shadow gate. The one-region discrepancy illustrates the operational hazard of relying on a snapshot illumination product for mission-critical site selection. The PSR-only pipeline was accordingly adopted as the standard for all downstream scoring.

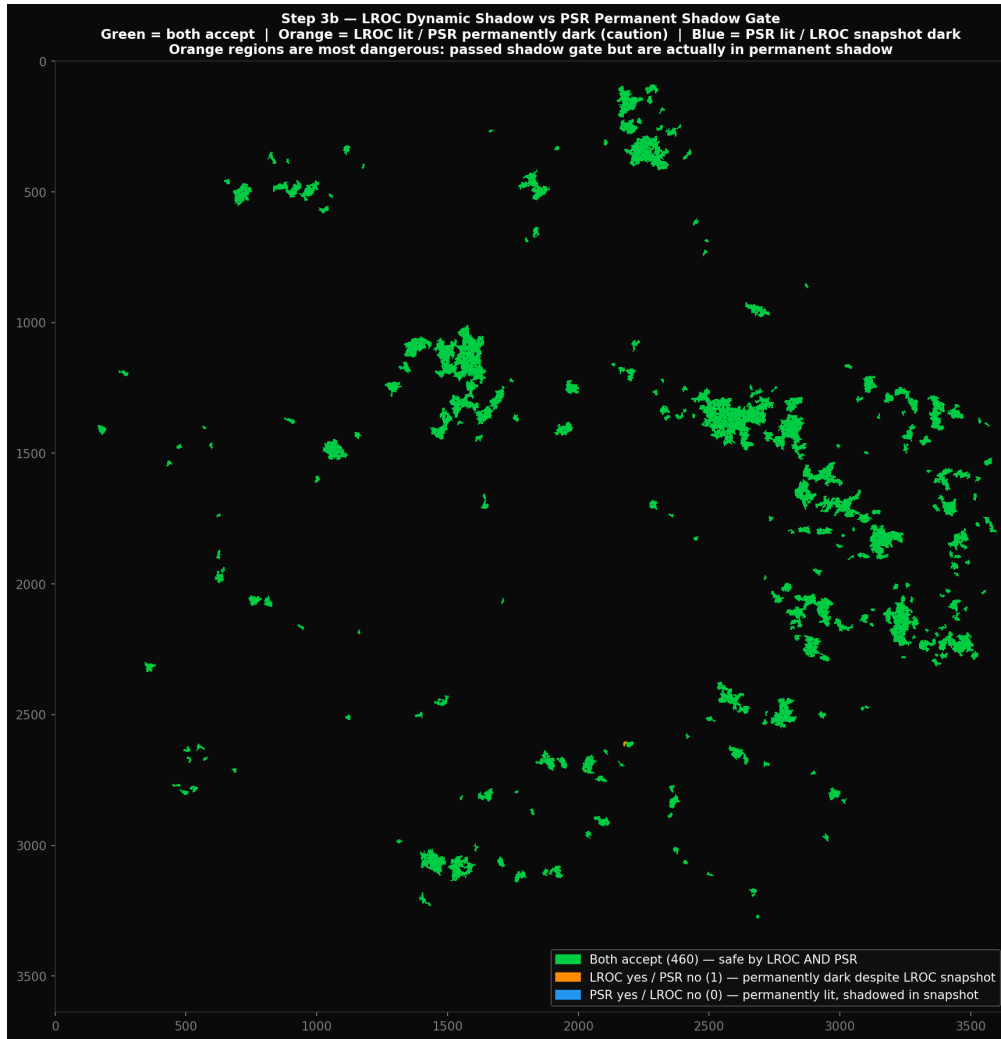


Figure 8.1: Step 3b validation: cross-check of the LROC dynamic shadow gate vs the PSR permanent-shadow gate. Green = both gates accept (460 regions); Orange = LROC accepts but PSR rejects (1 region, permanently dark despite appearing illuminated in the LROC snapshot); Blue = PSR accepts but LROC rejects (0 regions). The single orange region demonstrates the risk of snapshot-based shadow gating.

8.5 Science Score Distribution Check

What was checked: Whether the Science_Composite score distribution is consistent with a genuine, low-signal measurement environment rather than a scoring artefact.

How it was checked: The histogram of Science_Composite over all 1,285 QUALIFIED regions was examined (Figure 9.2C). The mean (0.076), P90 (0.124), and max (0.307) were recorded. The ECC distribution was separately tabulated.

Result: The distribution is right-skewed: the majority of regions cluster at low composite scores, with a long tail towards the high end. Six regions reach $ECC \geq 4$ (HIGH_CONVERGENCE). This pattern is consistent with the known sparsity of strong multi-instrument volatile signatures in the literature.

What it means: The scoring is not artificially inflated. The observed skewness reflects the genuine sparsity of detectable spectral absorptions at M3’s 280 m resolution near PSR boundaries, compounded by the partial radar coverage that forces many regions to rely on a single radar channel rather than the coalesced two-instrument score.

8.6 Weight-Sensitivity Analysis (Step 13)

What was checked: Whether the top-10 ranking is stable under perturbations of the weight vector $\mathbf{w}$.

How it was checked: Six weight sets (T0–T5, Table 7.2) were applied to all 1,285 QUALIFIED regions. For each weight set, the Science_Composite was recomputed and a new ECC was derived from the within-channel terciles of the rescored values. The top-10 under each weight set was then compared to the T0 baseline top-10 using a set-overlap count (0–10). ECC counts for ≥ 4 channels were also recorded under each weight set.

Result: Results are shown in Figure 8.2 and Table 8.1. Top-10 overlap with T0 ranges from 3 (T2: Radar-heavy) to 10 (T0 baseline, by definition). The equal-weight scheme (T5) achieves the highest overlap among the non-baseline sets (6/10, meeting the “robust” threshold). Three weight sets (T1, T3, T4) are “sensitive” ($< 6/10$). Notably, the six HIGH_CONVERGENCE regions are invariant across all six weight sets because ECC is derived from within-channel ranks rather than weighted scores.

Table 8.1: Sensitivity analysis results: top-10 overlap with T0 baseline

Weight set	Mean score	Median score	Top-10 overlap	Stability class
T0: Baseline	0.076	0.069	10/10	— (baseline)
T1: ICE-heavy	0.057	0.049	7/10	Robust
T2: Radar-heavy	0.102	0.095	3/10	Sensitive
T3: OH-heavy	0.066	0.053	4/10	Sensitive
T4: Mineral-heavy	0.052	0.045	4/10	Sensitive
T5: Equal	0.062	0.054	6/10	Robust

What it means: The Science_Composite ranking is most sensitive to the radar weight. When radar dominates (T2), the ranking shifts substantially because radar coverage is spatially heterogeneous across the study area: regions with both DFSAR and Mini-RF observations can receive a higher coalesced radar score than regions covered by only one instrument, which inflates their composite score relative to the baseline. The ECC-first ranking rule (Equation 7.10) partially mitigates this effect by placing the six HIGH_CONVERGENCE regions at the top of the list regardless of which weight scheme

is applied. The equal-weight scheme is the most robust non-baseline configuration and can be used as a secondary ranking for independent cross-comparison.

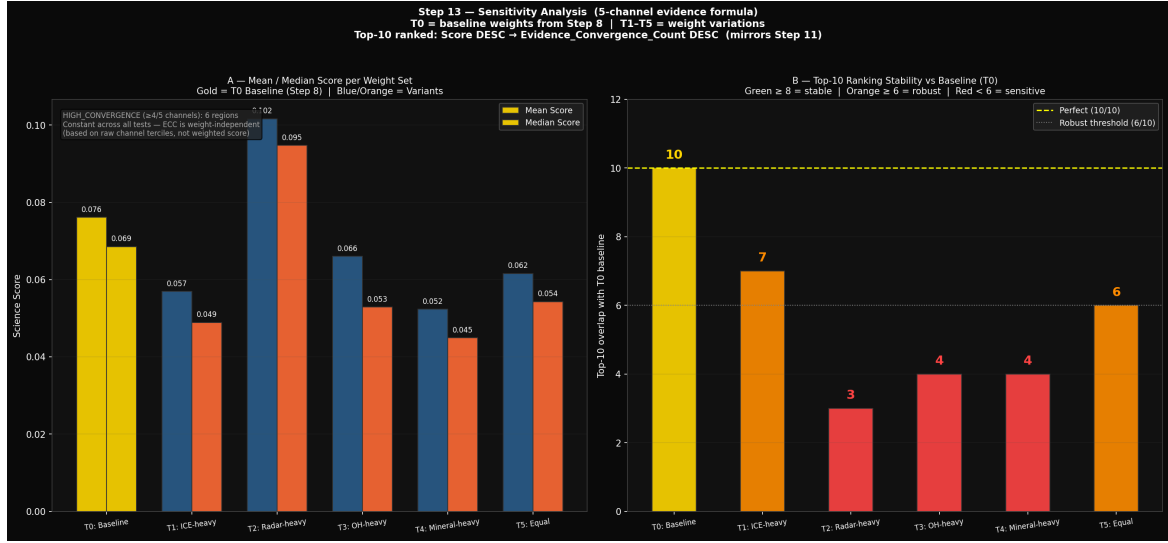


Figure 8.2: Step 13 sensitivity analysis. (A) Mean and median Science_Composite under each weight set. T2 (Radar-heavy) inflates absolute scores substantially; T4 (Mineral-heavy) deflates them. (B) Top-10 overlap count with the T0 baseline. Green ≥ 8 = stable; Orange ≥ 6 = robust; Red < 6 = sensitive. Only T1 (ICE-heavy, 7/10) and T5 (Equal, 6/10) meet the robust threshold; T2, T3, and T4 are sensitive. The six HIGH_CONVERGENCE regions are constant across all tests (shown in text box, panel A).

Chapter 9

Results

9.1 Input Layer Outputs

The dataset-ingestion step (Step 1) produced a machine-readable manifest of all source rasters (`file_description_table.csv`, 13 rows). Eleven rasters were registered to the $3,639 \times 3,639$, 25 m polar-stereographic grid. Native resolutions span more than three orders of magnitude: from 5 m for the LROC imagery ($18,195 \times 18,195$ pixels before resampling) to approximately 10 km for the GRAIL gravity product (9×9 pixels). The M3 mosaic contributes 84 spectral bands at 280 m resolution (325×325 pixels), of which 8 are carried forward. The PSR illumination raster covers the full $3,639 \times 3,639$ grid at 25 m and classifies 1,349,631 pixels (843.5 km^2 , 93.5% of the inner region of interest) as permanently shadowed.

9.2 Candidate Region Segmentation (Step 2)

Marker-controlled watershed segmentation of the LOLA slope raster produced 26,237 spatially exclusive regions covering $6,503.9 \text{ km}^2$, or 78.6% of the study grid (Figure 6.1). The region-size distribution is strongly right-skewed: the median region area is approximately 0.03 km^2 , which falls below the M3 footprint and confirms that the minimum-area gate in Step 3b will be the dominant removal criterion in practice. Importantly, each region is bounded by slope ridgelines and therefore respects natural terrain divides rather than the arbitrary boundaries of a rectangular grid.

9.3 PSR Map and Illumination (Step 2b)

The PSR binary map at 25 m (Figure 6.2A) shows that permanently shadowed terrain is concentrated in the interiors of large crater floors and is discontinuous near the pole, consistent with the topographic ray-tracing results reported in [3]. The log-illumination panel (Figure 6.2B) reveals that the annual illumination fraction varies by roughly four orders of magnitude across the covered area, from approximately 10^{-5} in the deepest crater cores to around $10^{-2.5}$ at illuminated patches immediately adjacent to crater rims.

9.4 Qualification Outcome (Step 3b)

Applying the four-block qualification gate to the 26,237 candidate regions produced the following outcome:

- **1,285 QUALIFIED** regions (4.9% of candidates; total area 521.4 km²; mean 0.406 km²; median 0.252 km²; range 0.079–3.511 km²). These form the analysis pool for all subsequent steps.
- **504 UNCERTAIN** regions (1.9%): flagged for rough terrain (271), high slope variability (193), or insufficient radar coverage (40).
- **24,448 REMOVED** regions (93.2%): predominantly below the M3 minimum mapping unit (21,785, 89.1% of all removals), with the remainder failing slope (2,301) or PSR-illumination (359) criteria (Table 6.1).

Within the QUALIFIED set, 1,242 of 1,285 regions (96.7%) have `psr_fraction` $\leq$ 0.10, placing them in a comfortable range of solar access by any operational standard (Figure 6.3C). Only 24 regions fall in the 0.30–0.50 band near the qualification ceiling, confirming that the gate is discriminating without being overly restrictive.

The spatial distribution of QUALIFIED regions is consistent with the expected geography of PSR-free terrain: they occupy illuminated inter-crater arcs around the pole, precisely the locations described in [3].

9.5 PSR Situational Context (Step 5)

Figure 9.1 shows two context quantities computed for the 1,285 QUALIFIED regions: PSR fraction per region (panel A) and distance to the nearest PSR boundary (panel B). Of the QUALIFIED pool, 256 regions lie within 2 km of a PSR boundary. These represent prime science-proximity candidates, because a rover stationed in the illuminated

zone could reach PSR-adjacent regolith through short sorties without returning to a distant charging station. These proximity values are retained as context attributes in the output tables but are *not* incorporated into the Science_Composite score, in order to maintain a clean separation between the operability gate (which uses PSR fraction as a power constraint) and the science ranking.

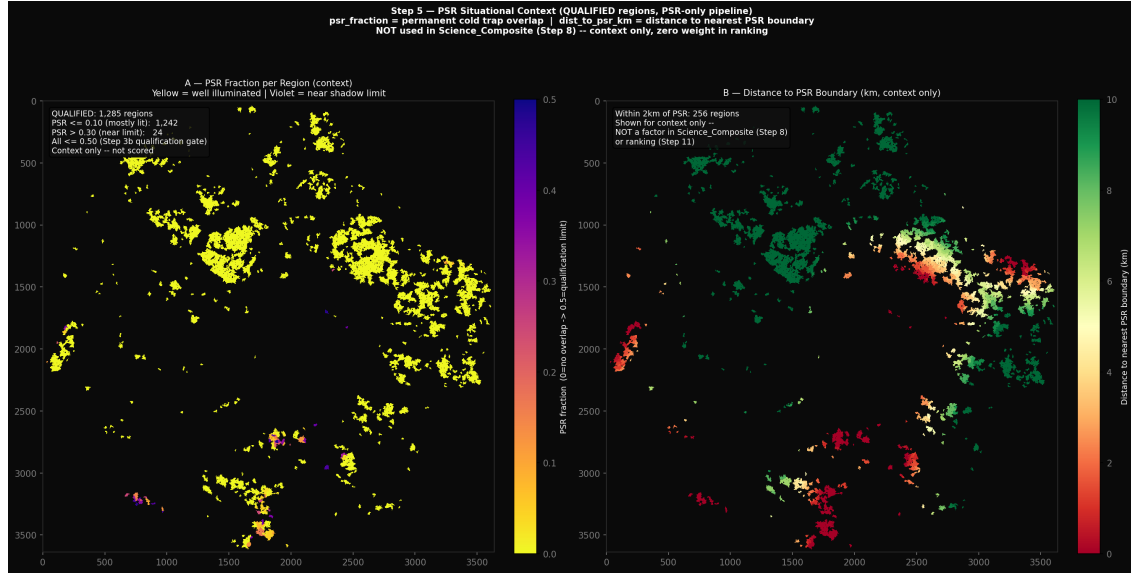


Figure 9.1: Step 5: PSR situational context for the 1,285 QUALIFIED regions. (A) PSR fraction: yellow = fully illuminated (1,242 regions with $\leq 10\%$ PSR); violet = near the 0.50 qualification limit (24 regions). (B) Distance to nearest PSR boundary (km): green = far from any cold trap; red-dark = within 2 km (256 regions). These quantities are context-only; they do not appear in Science_Composite.

9.6 Radar Feature Summary (Step 6)

Of the 1,285 QUALIFIED regions, radar coverage breaks down across three categories: BOTH DFSAR and Mini-RF available for a meaningful fraction, CPR_ONLY for regions outside the Mini-RF swath, and S1_ONLY for regions not covered by DFSAR. No QUALIFIED region has NEITHER radar source, which was guaranteed by the Step 3b UNCERTAIN flag acting as a filter. The CPR class distribution across QUALIFIED regions is dominated by `normal_low` (< 0.5) and `typical` (0.5 – 1.0) categories. A small number of regions fall in the `anomalous_high` (> 1.0) class, which the literature associates with either surface roughness or possible subsurface volume scattering from icy material [4].

9.7 Science Score and Composite Map (Step 8)

Figure 9.2 shows the Step 8 outputs for all 1,285 QUALIFIED regions. The Science_Composite panel (A) reveals that the highest composite scores are concentrated in several arc-like clusters of illuminated terrain, with individual bright spots corresponding to regions that show simultaneous spectral and radar enhancement above background levels. The Evidence_Convergence_Count panel (B) provides a complementary view based on multi-instrument agreement: only a small subset of regions achieves ECC = 4 or 5, and these are not necessarily the same regions as those with the highest weighted composite, because ECC is derived from within-channel ranks rather than weighted sums.

Score statistics over the 1,285 QUALIFIED regions: mean = 0.076, median = 0.069, P90 = 0.124, max = 0.307. The score distribution (Figure 9.2C) is right-skewed, consistent with the expected scarcity of strong multi-channel volatile signatures at 280 m resolution near the south pole.

Six regions reach $ECC \geq 4$ (HIGH_CONVERGENCE); 331 reach MODERATE_CONVERGENCE ($ECC = 2-3$); the remaining 948 are LOW_CONVERGENCE ($ECC \leq 1$).

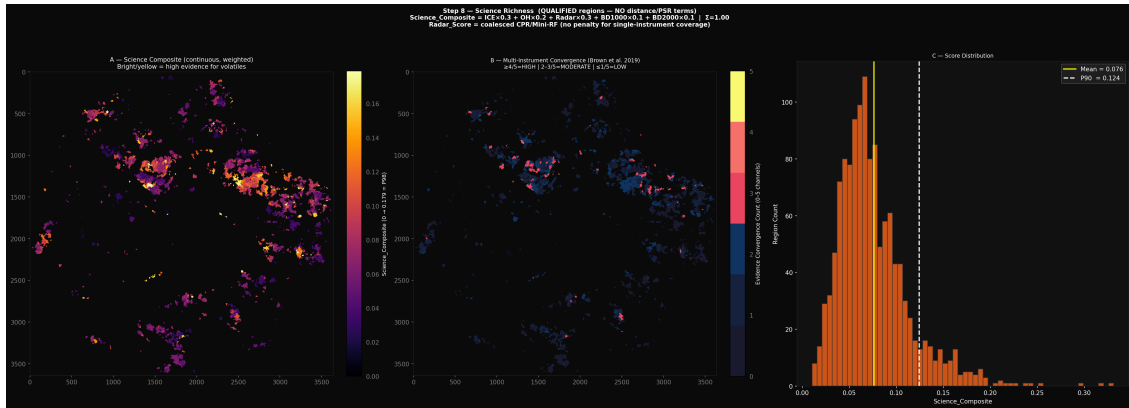


Figure 9.2: Step 8 Science Richness outputs for 1,285 QUALIFIED regions. (A) Science_Composite (continuous, weighted; bright/yellow = highest evidence). (B) Evidence_Convergence_Count (multi-instrument agreement; yellow = HIGH, ≥ 4 channels). (C) Score distribution histogram; mean = 0.076 (yellow line), P90 = 0.124 (dashed).

9.8 Feasible Target Pool (Step 10)

After applying the full operability gate ($\text{slope_mean} \leq 20^\circ$, $\text{slope_max} \leq 30^\circ$, $\text{pct_gt25} \leq 10\%$, $\text{psr_fraction} \leq 50\%$), the pool of 1,285 QUALIFIED regions is confirmed as the rover-accessible candidate set. Figure 9.3 maps this pool against the full study area, illustrating how accessible terrain is concentrated in narrow illuminated corridors

between the large PSR-filled craters that dominate the south-polar landscape.



Figure 9.3: Step 10: The 1,285 QUALIFIED regions (green) constituting the rover-accessible candidate pool, after operability gates. Regions are concentrated along illuminated inter-crater corridors.

9.9 Science-Priority Ranking (Step 11)

Figure 9.4 maps the top 20 science-priority targets (ranks 1–20) overlaid on the full QUALIFIED pool, coloured by Science_Composite. All six HIGH_CONVERGENCE regions appear in the top six positions (ranks 1–6), with ranks 7–20 occupied by MODERATE_CONVERGENCE (ECC = 3) regions. Table 9.1 lists the key attributes of these 20 regions.

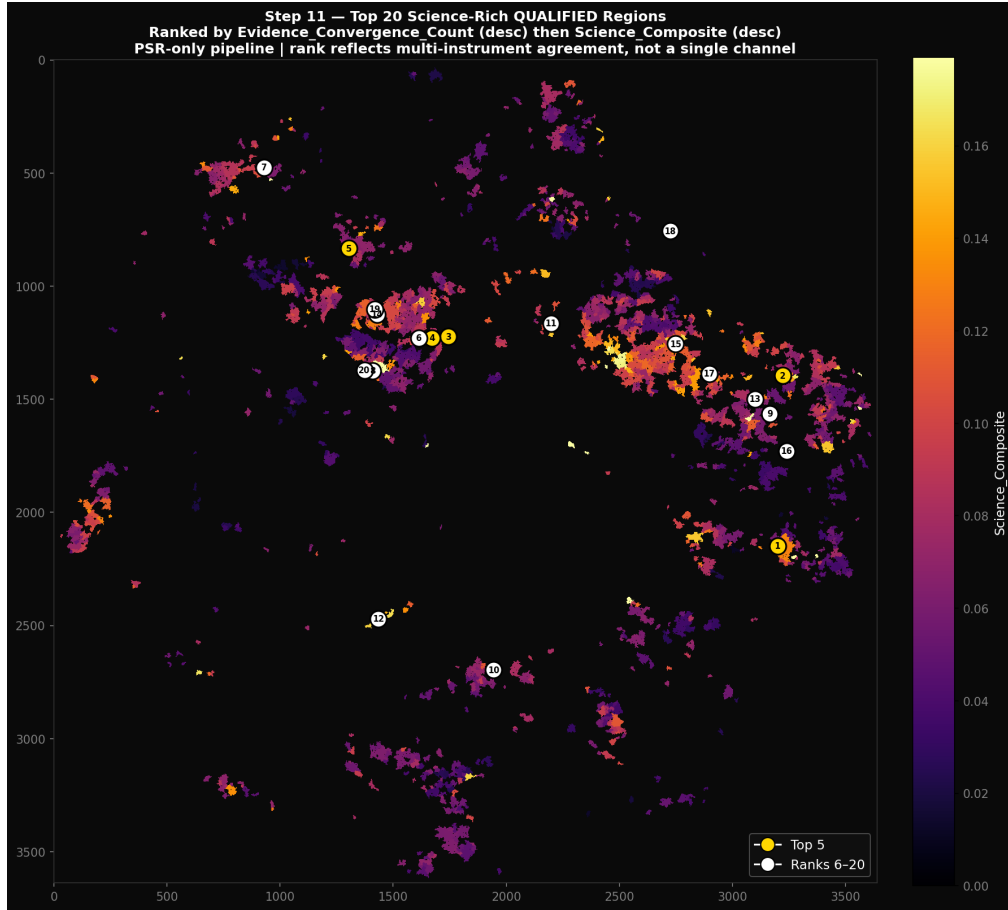


Figure 9.4: Step 11: Top 20 science-rich QUALIFIED regions ranked by ECC (descending) then Science_Composite (descending). Gold circles = top 5; white circles = ranks 6–20. Colour scale = Science_Composite. Ranks 1–6 are the six HIGH_CONVERGENCE ($\geq 4/5$ channels) regions.

Table 9.1: Top 20 science-priority targets (Step 11)

Rank	Region ID	Sci. Comp.	ECC	Band	Slope (°)	Area (km ²)	Dist. (km)
1	15598	0.209	4	HIGH	9.51	0.261	14.94
2	9163	0.193	4	HIGH	6.82	0.192	2.10
3	7446	0.191	4	HIGH	6.61	0.104	9.28
4	7454	0.178	4	HIGH	4.99	0.085	11.06
5	4071	0.158	4	HIGH	9.95	0.353	23.05
6	7584	0.130	4	HIGH	8.05	0.214	12.21
7	1750	0.226	3	MOD	10.19	0.146	25.03
8	8991	0.219	3	MOD	4.46	0.306	15.48
9	10987	0.193	3	MOD	5.86	0.126	6.64
10	19964	0.192	3	MOD	5.69	0.281	0.00
11	6912	0.192	3	MOD	7.57	0.083	4.38
12	17909	0.190	3	MOD	8.13	0.139	10.53
13	10385	0.187	3	MOD	6.86	0.103	5.80
14	6565	0.179	3	MOD	6.17	0.477	17.37
15	7687	0.174	3	MOD	5.24	0.116	6.46
16	12486	0.172	3	MOD	6.30	0.504	9.98
17	9162	0.170	3	MOD	6.12	0.159	6.58
18	3628	0.166	3	MOD	8.73	0.114	8.16
19	6375	0.165	3	MOD	7.94	0.166	17.97
20	8949	0.162	3	MOD	9.05	0.784	15.92

Note: ECC = Evidence Convergence Count. “HIGH” = HIGH_CONVERGENCE ($\geq 4/5$ channels in top tercile). “MOD” = MODERATE_CONVERGENCE. Dist PSR = distance to nearest PSR boundary (km); 0.00 for region 19964 indicates PSR immediately adjacent.

Among the six HIGH_CONVERGENCE regions, ranks 4 and 6 (Region 7454 and Region 7584) merit specific comment, as they represent the lower end of the Science_Composite range within this invariant top-6 set yet still satisfy every operational and scientific gate comfortably. Both regions have ECC = 4, with all four spectral/radar channels (ICE, OH, BD1000/BD2000, and DFSAR CPR) registering in the top tercile – i.e. full 4/4 `n_valid_spectral` coverage, so their composite scores rest on a complete evidence base rather than a partial one. Both also clear the rover-accessibility gate with margin: Region 7454 has a mean slope of 4.99° (max 19.21°) and a mean roughness of 4.54 m, while Region 7584 has a mean slope of 8.05° (max 18.44°) and a mean roughness of 6.49 m – both well inside the `HARD_SLOPE_MAX` = 20° and `MAX_ROUGH_MEAN` = 10.0 m thresholds. Neither region lies inside a PSR (`psr_fraction` = 0), so both are fully illuminated and pose no solar-power risk, while still sitting within 11–12 km of a PSR boundary (11.06 km and 12.21 km respectively), keeping a volatile-rich traverse target within feasible roving distance. Their candidate-region areas (0.085 km² and 0.214 km²) are also large enough to accommodate a landing ellipse and short-range survey path. Taken together, these properties confirm that

ranks 4 and 6 are not borderline inclusions on score alone, but genuinely *operationally safe, scientifically well-evidenced* candidates – consistent with the Step 13 finding that the top-6 HIGH_CONVERGENCE set is invariant across all weighting schemes.

9.10 Method Comparison (PSR vs LROC Shadow Gate)

The PSR-only gate (Step 3b) admitted 1,285 QUALIFIED regions compared to only 461 under the earlier LROC shadow gate. The additional regions that pass under the PSR gate are those whose single-epoch LROC shadow fraction exceeded the 0.50 threshold at the time of the image but whose annual PSR fraction is well below it, that is, terrain that was photographed in shadow at a particular solar geometry but is in fact sunlit for most of the lunar year. More critically, the one orange region in Figure 8.1 demonstrates the converse failure: the LROC gate admitted a site that is actually permanently shadowed (annual illumination < 0.001), and that would be operationally unsafe for a solar-powered rover throughout the year. This single discrepancy is sufficient to confirm that the PSR-based gate is the physically appropriate choice for solar-power-constrained mission planning.

9.11 Sensitivity Analysis Results (Step 13)

The sensitivity analysis results are summarised in Table 8.1 and Figure 8.2. The most important finding is that the six HIGH_CONVERGENCE regions (ranks 1–6) are *invariant* across all six weight sets, because ECC is computed from within-channel tercile thresholds rather than from the weighted composite. This invariance provides a strong practical argument for using ECC as the primary ranking key: the six shortlisted regions are recommended regardless of which weighting philosophy a mission planner subscribes to. Beyond rank 6, the composite-score ranking is sensitive to the radar weight (T2 overlap with T0: 3/10) but more stable under ICE-heavy (T1: 7/10) and equal-weight (T5: 6/10) configurations.

Chapter 10

Discussion

10.1 What Worked Well

The region-based pipeline succeeded in resolving the central tension stated in the research question: terrain accessibility and science value were evaluated jointly, yet kept structurally independent, so that neither criterion could corrupt the other. Of all the design choices in the pipeline, the PSR-based operability gate is arguably the most consequential. By deriving the solar-power constraint from a time-integrated annual illumination model rather than a single LROC snapshot, the gate is both physically grounded and reproducible. The cross-check in Section 8.4 demonstrated this concretely: the snapshot-based gate admitted one region that is permanently shadowed, an error the PSR gate correctly prevents.

The Evidence Convergence Count (ECC) proved to be the most reliable output of the science-scoring step. Because ECC counts how many independent evidence channels independently place a region in their top one-third — and does so from within-channel tercile ranks rather than from any weighted sum, making it entirely independent of the weight vector. The practical consequence is that the six HIGH_CONVERGENCE regions ($ECC \geq 4$) occupy the top of the ranking under every one of the six weight configurations tested. No plausible adjustment to the weights can displace these regions unless the underlying instrument data change. This is a meaningful property for mission planners who need a shortlist that will not be overturned by a retrospective argument about relative evidence weights.

The watershed segmentation on the LOLA slope raster produced terrain-bounded regions that respect natural ridgelines, giving a physically motivated unit for the aggregation of datasets with very different native resolutions. Using the M3 footprint ($280\text{ m} \times 280\text{ m} = 0.0784\text{ km}^2$) as the minimum mapping unit is similarly well-justified: it ties the lower size limit to the coarsest science instrument and avoids awarding scores

to regions that M3 cannot characterise independently.

10.2 What Did Not Work Well

The `Science_Composite` score distribution is strongly right-skewed, with 948 of 1,285 regions (73.7%) classified as `LOW_CONVERGENCE`. This is partly a genuine feature of the physical environment, given that strong multi-channel volatile signals are rare at these scales near the south pole, but it is also a consequence of the M3 footprint. At 280 m, M3 pixels are considerably larger than the median candidate region ($0.25 \text{ km}^2 \approx (0.5 \text{ km})^2$). Mixed-pixel effects reduce the apparent depth of any real absorption feature, because the spectral signal is averaged over terrain that may include both volatile-rich and bare-regolith material. The four M3-derived channels together carry 70% of the composite weight, so their limited spatial resolution directly constrains the ability of the composite score to differentiate sites at the 100–200 m scale.

The radar weight is the primary source of ranking instability beyond rank 6. When the radar weight is raised to 0.60 (T2: Radar-heavy), the top-10 overlap with the baseline falls to 3/10. This instability arises from a coverage asymmetry: regions with valid data from both DFSAR and Mini-RF receive a coalesced score that averages two independent observations, while `CPR_ONLY` or `S1_ONLY` regions receive a score from a single instrument. The coalescing equation (Equation 7.7) treats both instruments as equal contributors when both are available, but does not apply a coverage penalty to single-instrument regions. A future iteration of the pipeline could address this by incorporating a coverage-weight factor that discounts single-instrument radar scores proportionally.

Step 12 was retired because the original Pareto-front analysis comparing `Science_Composite` against slope became redundant once the Step 3b gate was in place: all `QUALIFIED` regions already satisfy $\text{slope_mean} \leq 20^\circ$, so a Pareto analysis over the 0–20° range that remains reveals no additional trade-off. Retiring the step rather than retaining it as an empty output reflects a deliberate design decision that is documented in the pipeline code.

10.3 Physical Meaningfulness

The spatial distribution of `HIGH_CONVERGENCE` regions is consistent with what the wider literature would lead one to expect for PSR-adjacent volatile deposits. The six regions at ranks 1–6 cluster in illuminated inter-crater terrain within a range of a few to tens of kilometres from PSR boundaries, which is precisely where spectral

OH/H₂O signatures have been reported to be strongest, specifically near areas that act as cold traps for thermally migrating volatiles [5]. Region 19964 (rank 10) has `dist_to_psr_km` = 0.00 km, placing it immediately adjacent to a PSR boundary. From a mission planning perspective, this is particularly attractive: a rover stationed in the illuminated zone could sample regolith at or near the cold-trap boundary in a short sortie, without committing to the sustained energy deficit of operating inside a shadowed crater.

That said, the spectral and radar indicators used in this pipeline are context indicators, not confirmed detections of water ice or any other volatile species. M3 cannot observe PSR interiors directly because the reflected solar flux inside those regions is insufficient to produce a usable signal, so all spectral evidence comes from illuminated terrain at the PSR margin rather than from within the cold trap itself. Elevated CPR values are similarly ambiguous: they may reflect surface roughness, blocky ejecta, or subsurface ice, and distinguishing among these interpretations requires either higher-resolution radar or in-situ measurement. These limitations are intrinsic to the available remote-sensing instruments and cannot be resolved within the scope of this analysis.

10.4 Limitations

1. **M3 resolution mismatch.** The 280 m M3 footprint is 3–10× larger than the median candidate region area. Mixed-pixel effects reduce the specificity of all four M3-derived spectral channels, limiting the ability of the composite score to resolve differences at scales below a few hundred metres.
2. **Radar coverage heterogeneity.** Not all regions have valid observations from both DFSAR and Mini-RF. Single-instrument radar scores are not directly comparable in absolute terms to dual-instrument coalesced scores, which creates a systematic advantage for better-covered regions when the radar weight is high.
3. **PSR model uncertainty.** The PSR threshold of $\text{illumination} < 10^{-3}$ follows Mazarico et al., but different illumination models based on alternative DEM products or different solar ephemeris ranges produce slightly different PSR masks. The 0.50 PSR-fraction gate provides some margin against this uncertainty, but it does not eliminate it.
4. **No ground truth.** There is no in-situ measurement of volatile abundance at any of the ranked candidate sites. The ranking reflects the convergence of remote-sensing evidence, not the confirmed presence of any specific resource.
5. **Static snapshot.** All inputs are either time-integrated or single-epoch products.

No seasonal variation or time-of-day dependence in the radar or spectral measurements is captured by the current pipeline.

10.5 Use in the Larger Digital Twin

The QUALIFIED region pool and the ranked shortlist produced by this pipeline are ready for direct consumption by the path-planning and confidence-classification layers of the digital twin. All output tables are keyed on `region_id` and `geo_x/geo_y`, which are the native join keys for every other layer registered to the $3,639 \times 3,639$ polar-stereographic grid. The ECC category (`evidence_band`) provides a discrete, three-level confidence tier that can be mapped directly to the trust levels used by the Intern 2 confidence-classification layer, without requiring that layer to reprocess the underlying spectral or radar rasters from scratch.

Chapter 11

Integration With the Digital Twin

Table 11.1 summarises the outputs produced in this project and their intended downstream use within the wider lunar south-pole digital twin. Each output is designed so that the consuming layer can read it directly from CSV or GeoTIFF without requiring access to the raw instrument rasters or any re-running of the upstream pipeline steps.

Table 11.1: Outputs and their downstream digital-twin uses

Output produced	Downstream use
<code>accepted_candidate_regions_psr_only.csv</code> (1,285 QUALIFIED regions with full attribute set)	Primary candidate set for the path-planning layer (Intern 2). Provides terrain statistics and PSR context for each reachable region without requiring re-processing of raw rasters.
<code>science_priority_ranked.csv</code> (ranked shortlist with Science_Composite and ECC)	Science-targeting layer input. Rank and ECC can be used directly as a priority queue for target selection in mission simulation.
<code>science_score_map.png</code> / <code>ranked_targets_map.png</code>	Visual layers for the digital-twin interface. Can be overlaid on the terrain DEM for spatial context during traverse planning.
evidence_band column ({HIGH, MODERATE, LOW}_CONVERGENCE)	Trust-tier input for the confidence-classification layer (Intern 2). Maps directly to a three-level confidence class without requiring spectral or radar reprocessing.
<code>radar_features.csv</code> (<code>radar_flag</code> , <code>dfsar_cpr_class</code>)	Instrument-coverage metadata for the confidence layer. Flags regions where radar evidence is absent or single-instrument, enabling the confidence layer to downweight those regions accordingly.
<code>grail_regional_context_only.csv</code>	Regional geophysical context for the integration layer. Not used in scoring but available for subsurface-structure interpretation in future work.
<code>constraint_sensitivity_results.csv</code>	Robustness documentation for the digital-twin provenance layer. Records which weight assumptions were tested and how the ranking responds to each perturbation.

Format and Interface Requirements

All output tables are comma-separated values (CSV) keyed on `region_id` (integer) and spatial coordinates (`geo_x`, `geo_y` in polar-stereographic metres). Any layer in the digital twin that uses the same $3,639 \times 3,639$, 25 m polar-stereographic grid can join to these tables by region label or by nearest-centroid spatial query, without any additional coordinate transformation.

The qualification raster (`accepted_candidate_regions_psr_only.tif`) encodes region labels as integer pixel values on the same 25 m grid. It can be overlaid directly on terrain, illumination, or radar rasters without reprojection, since it already shares the same coordinate reference system and pixel alignment.

Main Assumptions and Limitations

The outputs assume that the PSR illumination model used in Step 2b is accurate to within the margin provided by the 0.50 PSR-fraction gate. They also assume that M3 L2 reflectance values are properly calibrated and that the band-depth indicators are computed on a consistent spectral continuum. If the PSR model or the M3 calibration is updated, Steps 2b and 4 onwards would need to be re-run before the downstream layers are refreshed.

The ECC-based trust tier is computed from tercile thresholds derived from this specific pool of 1,285 QUALIFIED regions. If the pool changes (for example, because the operability gate is revised) the tercile thresholds will shift and all ECC assignments must be recomputed. This dependency is tracked explicitly in the pipeline code and is not hidden in the output tables, so that a downstream consumer can verify whether the ECC values remain valid for their version of the candidate pool.

Chapter 12

Conclusion

This report described a thirteen-step pipeline for identifying rover-accessible, science-priority candidate sites near the lunar south pole, built by fusing terrain, illumination, spectral, and radar datasets at the level of discrete, terrain-bounded regions. The work was motivated by the recognition that accessibility and science value cannot be evaluated in isolation: a site that is safe to reach but scientifically uninformative is of limited mission value, and a site that is scientifically compelling but operationally inaccessible is equally irrelevant. Six principal conclusions are drawn from the analysis.

1. The PSR-based operability gate is the critical structural choice. Replacing a single-epoch LROC shadow snapshot with a time-integrated annual illumination fraction (threshold $< 10^{-3}$, following Mazarico et al.) produced a gate that is both physically grounded and operationally conservative. The cross-check against the earlier snapshot-based gate found one region that the snapshot admitted but the PSR gate correctly excluded: a site that is permanently dark despite appearing momentarily illuminated in a single LROC frame. The PSR gate reduces 26,237 watershed-segmented candidates to 1,285 QUALIFIED regions (4.9%, 521.4 km²), all with `slope_mean` $\leq 20^\circ$ and `psr_fraction` ≤ 0.50 .

2. The Evidence Convergence Count is the most robust ranking output. The six HIGH_CONVERGENCE regions (`ECC` $\geq 4/5$ channels in their top tercile) occupy ranks 1–6 under every one of the six weight configurations tested. This invariance holds because ECC is derived from within-channel tercile ranks rather than from any weighted composite, making it entirely independent of the weight assignment. For mission planners, this means the primary shortlist is stable regardless of which weighting philosophy is applied.

3. The Science_Composite ranking is weight-sensitive beyond rank 6. The top-10 overlap with the baseline falls to 3/10 under a radar-dominant weight set and reaches a maximum of 7/10 under ICE-heavy weighting. The equal-weight scheme

achieves 6/10 overlap, the best performance among non-baseline configurations, and can serve as a secondary robustness check for planners wary of the assumptions embedded in the baseline weights.

4. M3 resolution is the principal limitation of the science score. The 280 m M3 footprint is 3–10× larger than the median candidate region ($\sim 0.25 \text{ km}^2$). Mixed-pixel effects dilute the spectral signal and limit the ability of all four M3-derived channels to resolve differences at the scale of individual candidate regions. M3 also cannot observe PSR interiors, so spectral evidence necessarily comes from illuminated terrain at the cold-trap margins. Higher-resolution hyperspectral or active illumination data would be needed to sharpen spectral discrimination at the 25–100 m scale.

5. All outputs are designed as direct inputs to the digital-twin pipeline. The QUALIFIED region table, the ranked shortlist, the `evidence_band` trust tier, and the qualification raster are all registered to the same 25 m polar-stereographic grid and can be consumed by the path-planning and confidence-classification layers without additional reprojection or data reprocessing. This interoperability was a design requirement from the outset and has been maintained throughout.

6. The suggested next step is field validation of the top-6 targets. The six HIGH_CONVERGENCE regions should be cross-checked against existing high-resolution LROC NAC imagery to confirm that their mean slope values are not artefacts of 25 m DEM resolution, and against any available Mini-RF or DFSAR full-resolution stacks to assess whether the elevated radar signal is spatially coherent within each region. Definitive confirmation of volatile presence at any of these sites would ultimately require a surface asset capable of collecting near-infrared spectra, neutron-flux measurements, or thermal-inertia observations in situ.

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Appendix A

Supplementary Material

A.1 Output File List

Table A.1: Complete output file list

File	Step	Description
file_description_table.csv	1	Manifest of all 11 input rasters with resolution, CRS, no-data values.
candidate_regions.csv	2	All 26,237 candidate regions with terrain, illumination, spectral and radar aggregates (51 columns).
shadow_fraction_by_region.csv	2	Shadow fraction statistics for all 26,237 regions.
candidate_regions_map.png	2	LOLA slope raster and 26,237 watershed regions coloured by mean slope.
psr_map.png	2b	Binary PSR mask and log-illumination panels.
candidate_region_qualification_psr_only.csv	3b	Full 26,237-row qualification table with status and reason.
accepted_candidate_regions_psr_only.csv	3b	1,285 QUALIFIED regions (master table for Steps 4–13).
uncertain_candidate_regions_psr_only.csv	3b	504 UNCERTAIN regions.
accepted_candidate_regions_map_psr_only.png	3b	Qualification map (green/orange/red/grey).

Table A.1: Complete output file list (continued)

File	Step	Description
psr_coverage_map.png	5	PSR fraction and distance-to-PSR context maps for QUALIFIED regions.
radar_features.csv	6	Radar CPR and Mini-RF statistics with coverage flags.
science_context_features.csv	7	Summary table of spectral, radar, and PSR context for QUALIFIED regions.
science_score.csv	8	Science_Composite, normalised channels, ECC, evidence_band for all 1,285 QUALIFIED regions.
science_score_map.png	8	Science_Composite and ECC maps with score distribution histogram.
feasible_target_regions.csv	10	QUALIFIED region pool (identical to Step 3b output, confirmed after operability gate).
science_priority_ranked.csv	11	1,285 QUALIFIED regions with rank, Science_Composite, ECC, and all attributes.
ranked_targets_map.png	11	Top-20 science-priority targets on Science_Composite background.
constraint_sensitivity_results.csv	13	Mean, median score and top-10 overlap for six weight sets.
sensitivity_summary.png	13	Sensitivity bar charts (mean/median score and top-10 overlap).

A.2 Code Execution Order

The entire pipeline is contained within a single script: `ALL_STEPS_-_FINAL_PIPELINE.py`. Steps execute sequentially in the order: 1, 2, 2b, 3b, 4, 5, 6, 7, 8, 10, 11, 13. Step 12 was retired (see Section 6).

Required Python packages: `numpy`, `pandas`, `matplotlib`, `scipy`, `rasterio`, `scikit-image` (watershed), `scikit-learn`.

Set the `DATA_DIR` and `OUT` path variables at the top of the script before execution. All outputs are written to `OUT`.

A.3 Method and Equation Summary

Table A.2: Method, equation, and algorithm summary

Step	Method / Equation	Purpose and reference
Step 2	Marker-controlled watershed	Terrain-bounded region segmentation [1, 2]
Step 2b	PSR threshold ($< 10^{-3}$)	Annual illumination classification [3]
Step 3b	Four-block decision gate	Operability qualification
Step 4	Band-depth formula (Eq. 7.1)	M3 spectral indicator extraction [7, 8]
Step 8	Min-max normalisation (Eq. 7.6)	Channel-scale equalisation
Step 8	CPR clip [0, 1.5] + normalise	Radar score preparation [4]
Step 8	Coalesced radar (Eq. 7.7)	Single radar evidence channel
Step 8	Science_Composite (Eq. 7.8)	Weighted 5-channel evidence score
Step 8	ECC tercile count (Eq. 7.9)	Weight-independent convergence metric [9]
Step 11	ECC-first sort (Eq. 7.10)	Final science-priority ranking
Step 13	Six weight sets (Table 7.2)	Sensitivity analysis